Exploratory Data Analysis on a Cars Dataset

Student Project Report

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Submission Date	19 October 2026
Dataset File	car data.csv (301 rows, 9 columns)
Dataset Source	**[Vehicle Dataset from CarDekho]**

Tools used: Python (pandas, NumPy, matplotlib, seaborn).

1. Objective

The purpose of this project is to carry out a complete Exploratory Data Analysis (EDA) on a used-vehicle dataset. The aim is to understand the structure of the data, check and improve its quality, find outliers, and identify patterns and relationships between the variables using statistics and visualisations. No machine-learning model is built.

The following questions were selected because the dataset contains the columns needed to answer them:

- How are selling prices distributed, and are there outliers?
- Which fuel types, transmission types and seller types are most common?
- How does the present price of a vehicle relate to its selling price?
- How do the manufacturing year (vehicle age) and kilometres driven relate to the selling price?
- Do automatic and manual vehicles differ in selling price?
- Which numerical features are strongly correlated with each other?

2. Dataset Overview

Source: : Vehicle Dataset from CarDekho

]. The file used is *car data.csv*. It contains **301 rows** and **9 columns** (before cleaning). No data dictionary came with the file, so the meaning of each column below is based on its name and the values it contains.

Column	Meaning (based on column name)	Data type	Variable type
Car_Name	Name / model of the vehicle	object (text)	Categorical
Year	Year of the vehicle (treated as manufacturing year)	int64	Numerical (discrete)
Selling_Price	Price at which the vehicle is being sold	float64	Numerical (continuous)
Present_Price	Present (new / showroom) price of the same model	float64	Numerical (continuous)
Kms_Driven	Distance already driven, in kilometres	int64	Numerical (continuous)
Fuel_Type	Fuel used: Petrol, Diesel or CNG	object (text)	Categorical
Seller_Type	Dealer or Individual	object (text)	Categorical
Transmission	Manual or Automatic	object (text)	Categorical
Owner	Number of previous owners	int64	Numerical (discrete count)

Note: the file does not state the currency or unit of the two price columns, so prices are reported in "dataset price units" throughout this report.

Sample records (first five rows of the cleaned dataset)

Car_ Name	Year	Selling_ Price	Present_ Price	Kms_ Driven	Fuel_ Type	Seller_ Type	Transmission	Owner
ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

3. Data Quality and Cleaning

The following checks were carried out on the raw file and the decision taken for each is explained.

Check	Finding	Decision / action
Missing values	0 missing values in all 9 columns.	No imputation or row removal needed.

Duplicate rows	2 fully duplicated rows (row indices 17 and 93; they repeat an *ertiga 2016* record and a *fortuner 2015* record).	Removed, keeping the first occurrence, so that the same listing is not counted twice.
Check	**Finding**	**Decision / action**
Data types	All columns already have suitable types (numbers stored as int64 / float64, text as object).	No type conversion needed. Owner is kept numeric because it is a count.
Category labels	Fuel_Type has 3 unique labels, Seller_Type 2 and Transmission 2, with no spelling or case variants.	No changes needed.
Car_Name text	11 records had extra leading, trailing or double spaces, and the capitalisation was mixed (for example, "KTM 390 Duke ").	Trimmed spaces and converted to lower case. Unique names reduced from 98 to 97.
Outliers (IQR rule)	Selling_Price: 16 values above 13.73. Present_Price: 14 above 22.80. Kms_Driven: 8 above 99,709.	**Kept.** The high prices belong to expensive models (e.g., land cruiser at 35.0). One record has 500,000 km (activa 3g); it is unusual but cannot be proven wrong, so it is kept and flagged.

Dataset size before and after cleaning

Measure	Before cleaning	After cleaning
Rows	301	299
Columns	9	9
Missing values	0	0
Duplicate rows	2	0
Unique Car_Name values	98	97

4. Descriptive Analysis

All statistics below are calculated on the cleaned dataset (299 rows).

Numerical columns

Column	Count	Mean	Median	Std. Dev.	Min	Max
Year	299	2013.62	2014	2.90	2003	2018
Selling_Price	299	4.59	3.51	4.98	0.10	35.00
Present_Price	299	7.54	6.10	8.57	0.32	92.60
Kms_Driven	299	36,917	32,000	39,015	500	500,000
Owner	299	0.04	0.00	0.25	0.00	3.00

The mean Selling_Price (4.59) is higher than the median (3.51) and the mean Present_Price (7.54) is higher than its median (6.10), so both price columns are right-skewed (skewness of Selling_Price = 2.54). Kms_Driven has a very large maximum (500,000 km) compared with its median (32,000 km). Years run from 2003 to 2018, with a median of 2014.

Categorical columns

Column	Category counts (percentage of 299 rows)
Fuel_Type	Petrol: 239 (79.9%); Diesel: 58 (19.4%); CNG: 2 (0.7%)
Transmission	Manual: 260 (87.0%); Automatic: 39 (13.0%)
Seller_Type	Dealer: 193 (64.5%); Individual: 106 (35.5%)
Owner (previous owners)	0: 288 (96.3%); 1: 10 (3.3%); 3: 1 (0.3%)
Car_Name (97 unique)	Most frequent: city: 26; corolla altis: 16; verna: 14; fortuner: 10; brio: 10

Average selling price by group

Group	Count	Mean Selling_Price	Median Selling_Price
CNG	2	3.10	3.10
Diesel	58	10.10	7.60
Petrol	239	3.26	2.65
Automatic	39	9.07	5.75
Manual	260	3.92	3.25
Dealer	193	6.63	5.25
Individual	106	0.87	0.52

The CNG group has only 2 records and the Owner = 3 group only 1, so their statistics should not be generalised.

5. Visual Analysis

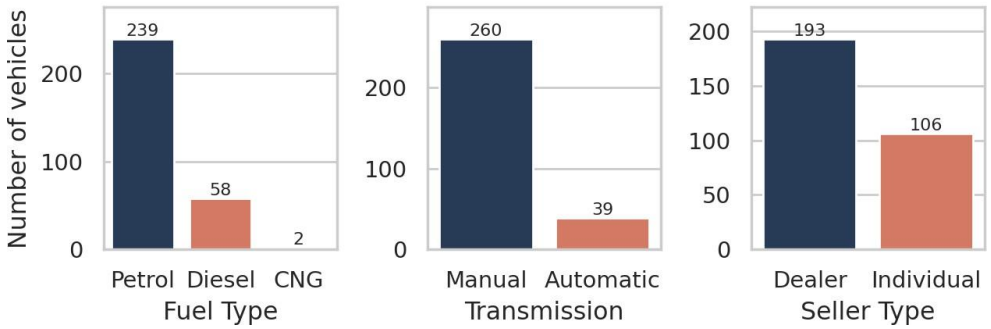
Eight charts of six different types (histogram, bar / count plot, box plot, scatter plot, line chart and heatmap) are shown below. Each chart is followed by a short observation based on what the chart shows.

Figure 1: Distribution of Selling Price



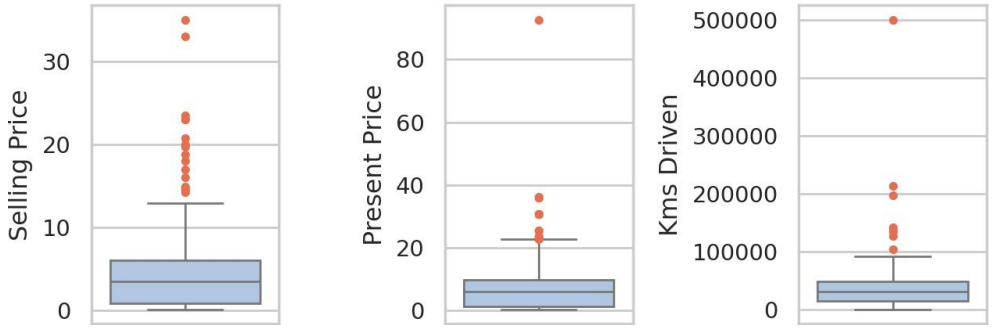
Observation: The distribution is strongly right-skewed. The tallest bar is the lowest price band, and 78 vehicles have a Selling_Price below 1. The mean (4.59) lies to the right of the median (3.51) because a few expensive vehicles (up to 35) stretch the tail.

Figure 2: Count of Vehicles by Fuel Type, Transmission and Seller Type



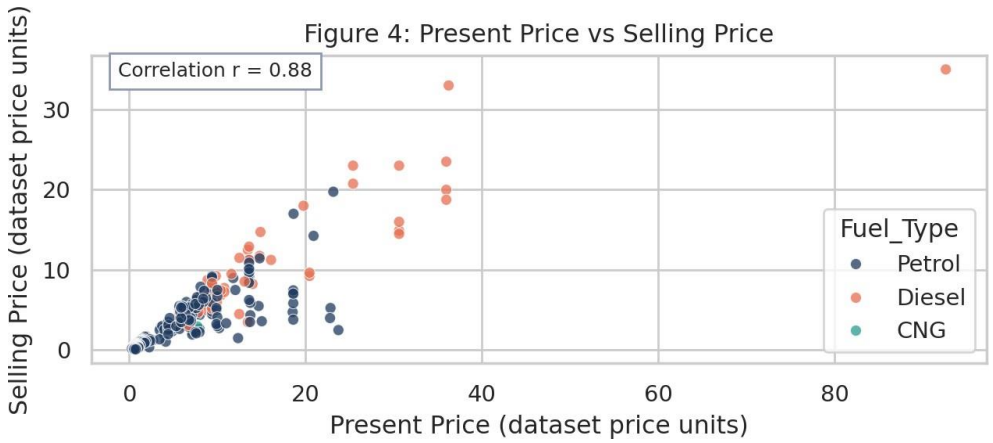
Observation: Petrol is by far the most common fuel (239 of 299, 79.9%), followed by Diesel (58); CNG appears only 2 times. Manual transmission dominates (260 vs 39 automatic), and Dealers (193) list more vehicles than Individuals (106).

Figure 3: Box Plots Used to Identify Outliers



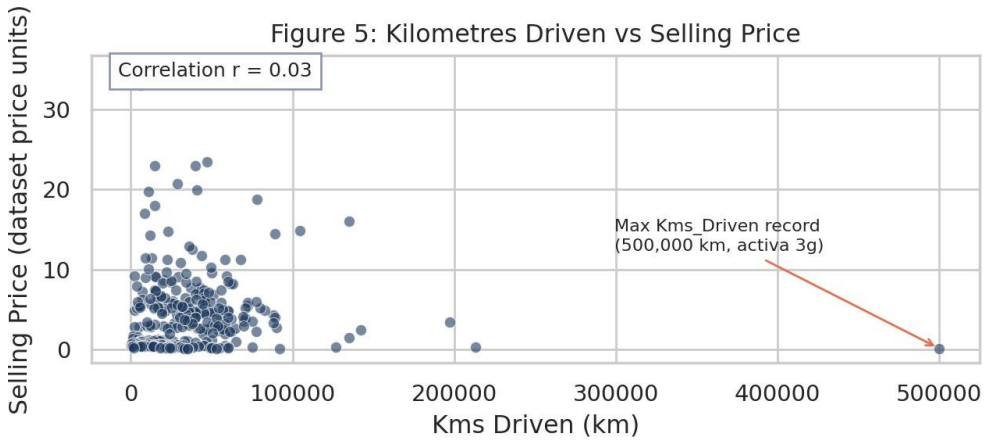
Observation: All three variables have points beyond the whiskers. By the IQR rule there are 16 high outliers in Selling_Price, 14 in Present_Price and 8 in Kms_Driven. The most extreme points are a Selling_Price of 35 (land cruiser), a Present_Price of 92.6 (land cruiser) and 500,000 km (activa 3g).

Figure 4: Present Price vs Selling Price



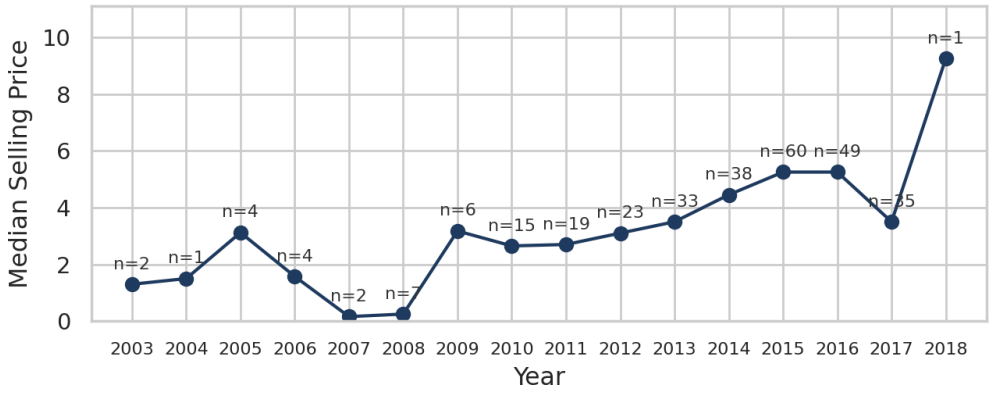
Observation: Selling_Price rises almost in a straight line with Present_Price (r = 0.88). Diesel vehicles (orange) tend to occupy the higher-priced upper right part. Some points fall well below the trend: 9 of the 12 Petrol vehicles with Present_Price of 18 or more sold for less than 8.

Figure 5: Kilometres Driven vs Selling Price



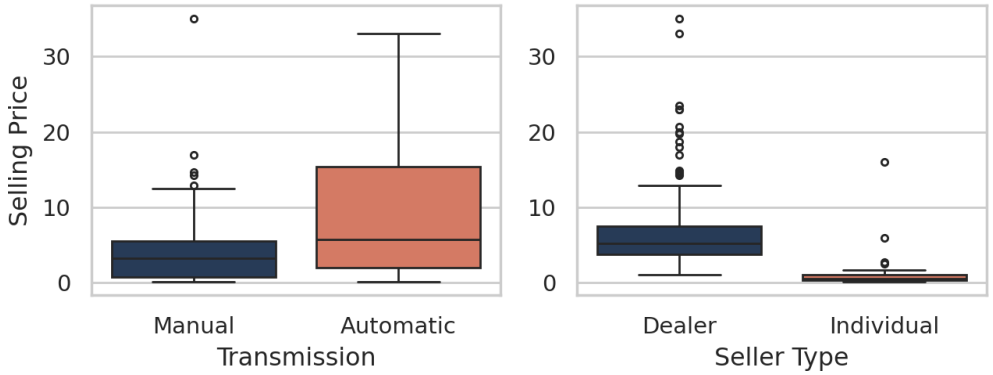
Observation: There is no clear pattern between Kms_Driven and Selling_Price (r = 0.03). 97.3% of vehicles have under 100,000 km, and they span the whole price range. All 6 vehicles priced above 20 have 78,000 km or fewer, and the 8 records above 100,000 km all sell for 16 or less. One extreme record (500,000 km, activa 3g, price 0.17) stands far from the rest.

Figure 6: Median Selling Price by Manufacturing Year



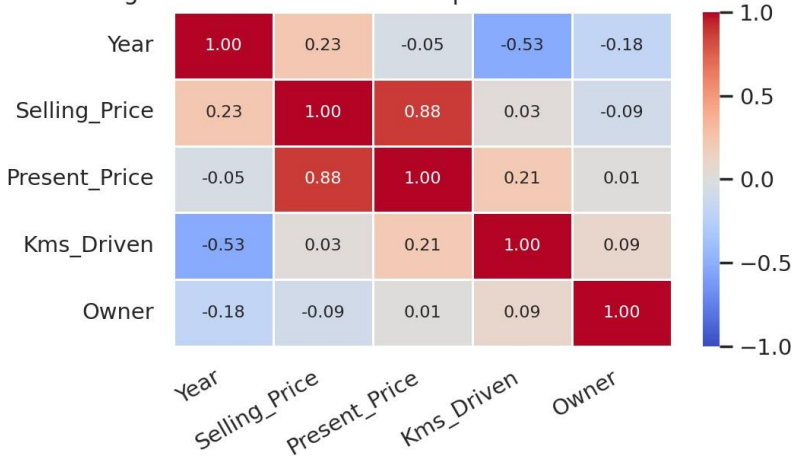
Observation: Median price is generally higher for newer vehicles, rising from 2.65 (2010) to 5.25 (2015 and 2016). It then falls to 3.50 for 2017. The line is uneven in the early years and for 2018 because those years have very few records (n = 1 for 2018), so those points are not reliable.

Figure 7: Selling Price by Transmission and by Seller Type



Observation: Automatic vehicles have a higher median Selling_Price (5.75) than manual ones (3.25) and a much wider spread. Dealer listings have a median of 5.25 against 0.52 for Individuals. These differences are partly caused by the mix of vehicle types in the data (see Section 6, Finding 5).

Figure 8: Correlation Heatmap of Numerical Columns



Observation: Selling_Price and Present_Price are strongly positively correlated (0.88). Year and Kms_Driven are moderately negatively correlated (-0.53), meaning newer vehicles have fewer kilometres. Year has a weak positive link with Selling_Price (0.23), while Kms_Driven (0.03) and Owner (-0.09) show almost no linear relationship with it. Correlation does not prove cause and effect.

6. Key Findings

1. **Present_Price is the strongest indicator of Selling_Price.** Their correlation is 0.88. A vehicle typically sells for about 65% of its Present_Price (median ratio 0.65, mean 0.63); in no record does the Selling_Price exceed the Present_Price (highest ratio 0.99).
2. **Selling prices are right-skewed with plausible high-value outliers.** Mean 4.59 vs median 3.51, skewness 2.54. 78 vehicles sell below 1, while 16 vehicles lie above the IQR limit of 13.73 (e.g., land cruiser at 35). These were kept because they are plausible records.
3. **The market in this dataset is dominated by petrol, manual, dealer-listed vehicles.** Petrol is 79.9%, manual is 87.0%, dealers are 64.5%, and 96.3% of vehicles have had no previous owner (Owner = 0).
4. **Newer vehicles have fewer kilometres, but kilometres alone do not explain price.** Year and Kms_Driven have r = -0.53, whereas Kms_Driven and Selling_Price have r = 0.03. Year has a weak positive link with price (r = 0.23), which suggests that in this data price is linked far more to the type of vehicle (its Present_Price) than to age or mileage.
5. **The dataset mixes cars with two-wheelers, which strongly affects group comparisons.** Based on the names in Car_Name (a keyword check, e.g., Royal Enfield, Bajaj, Honda Activa), about 101 of 299 records (33.8%) look like motorcycles or scooters. Their median Selling_Price is 0.50 against 5.25 for the rest, and all of them are listed by Individual sellers (Individual sellers: 106 records in total; only 5 are cars). The large Dealer vs Individual price gap therefore mostly reflects vehicle type, not the seller type itself.
6. **Automatic and diesel vehicles sell for more than manual and petrol ones.** Median Selling_Price: Automatic 5.75 vs Manual 3.25; Diesel 7.60 vs Petrol 2.65. The gap remains when only the car-type records are compared (Automatic 7.62 vs Manual 4.93; Diesel 7.60 vs Petrol 4.70), but these are descriptive differences only, because model, size and segment are not controlled.

7. Limitations

- **Unknown source and meaning of columns.** There is no data dictionary, so units (currency) and definitions of Year and Present_Price are inferred from the column names.
- **Mixed vehicle types.** The data has no vehicle-type column, although many records are two-wheelers. The two-wheeler count in this report comes from a keyword check on Car_Name and is approximate.
- **Limited size and uneven groups.** Only 299 records remain; CNG has 2 records, Owner = 3 has 1, and Year 2018 has 1. Results for such groups are not reliable.
- **Missing explanatory features.** Engine size, horsepower, mileage (km/l), body type, location, condition and brand are not available, so many price differences cannot be explained.
- **Unverified extreme values.** The record with 500,000 km could be a data-entry error; it was retained and not confirmed. Two duplicate rows were removed, although they could in principle be separate listings.
- **Correlation is not causation.** The analysis is descriptive. Relationships between variables may be influenced by other factors not present in the data.

8. Conclusion and Future Scope

This EDA examined a used-vehicle dataset of 301 raw records, which became 299 records after removing 2 duplicate rows and standardising Car_Name text. No missing values were found. The main result is that Selling_Price is closely tied to Present_Price (r = 0.88), is right-skewed, and has high-value outliers that appear to be genuine. Kilometres driven and number of owners show little linear relationship with price, and manufacturing year only a weak one. Automatic and diesel vehicles have higher median prices, but the presence of two-wheelers in the data (about 33.8% of records by name) affects comparisons such as Dealer vs Individual.

Challenges faced: deciding how to treat outliers that looked real rather than like errors, separating the effect of vehicle type from the effect of seller type or transmission, and working with very small groups.

Future scope:

- Obtain or add a vehicle-type column and brand column, and analyse cars and two-wheelers separately.
- Add features such as engine size, fuel efficiency, body type and location.

- Apply a log transformation to Selling_Price to reduce skewness, and build a regression model to predict price (not part of this EDA).
- Collect a larger and more balanced dataset, and confirm the source, units and meaning of each column.

Acknowledgement

Dataset: : Vehicle Dataset from CarDekho

]. Libraries: pandas, NumPy, matplotlib and seaborn. All statistics and charts in this report were generated from the file *car data.csv*.